

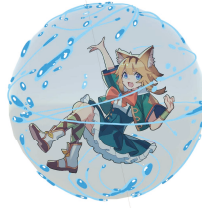
SpiCa: Stereoscopic Effect Design with 3D Pottery Wheel-type Transparent Canvas

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(a) Flame effect on a conical canvas



(b) Water effect on a spherical canvas



(c) Wind effect on a cylindrical canvas

Figure 1: Examples of flow effects drawn on transparent surface of revolution canvases used in SpiCa.

ABSTRACT

Flow effects such as flames, smoke, and liquids play an important role in activating illustrations, but drawing these effects requires artistic expertise as well as a great deal of effort. In this paper, we propose a method for adding stereoscopic flow effects to character illustrations using various shapes of 3D pottery wheel-type transparent canvases. One approach to designing a flow effect to decorate a character relies on simple curved geometry to beautify its flow in an organized composition. We extend this approach to present a drawing system—SpiCa (spinning canvas), which enables users to use transparent surface of revolution canvases to design 3D flow effects. User evaluations showed that users were able to create such effects more easily and effectively and reduce their workload with SpiCa in comparison with an existing 2D illustration tool.

CCS CONCEPTS

• Computing methodologies → Graphics systems and interfaces.

KEYWORDS

User Interface, Visual Effects, Illustration

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1 INTRODUCTION

Flow effects, which conceptualize amorphous phenomena such as light, flame, smoke, water, and wind, are essential in artworks in a wide range of applications. Such effects play an important role in activating these works and making characters' emotions and scene development more attractive. In general, traditional hand-drawn animation offers full artistic control and expression but requires significant expertise and manual labor. Physical simulation, however, may produce realistic expressions but lacks directability.

In recent years, sketch-based interfaces for designing animated flow effects have been explored [Kazi et al. 2014] [Xing et al. 2016]. However, using general sketch-based methods, only 2D strokes can be drawn, and thus, it is difficult to create complex 3D flow effects.

In this paper, with a focus on flow effects in character illustrations, we present a novel drawing system called SpiCa (spinning canvas) for creating stereoscopic flow effects with 3D pottery wheel-type transparent canvases. In character illustrations, the path of flow effects is often drawn along the surface of a simple shape to organize the composition. In SpiCa, such surfaces are realized as revolving canvases that surround characters, as shown in Fig. 1, and enable beginners to easily sketch 3D flow effects.

2 RELATED WORK

The computer graphics community has exploited physical simulation as a powerful tool for creating flow effects. Along with the progress of such methods, in contrast with the increase in the complexity of flow effects, controlling them have become a difficult task. As a fluid control method, Treuille et al. proposed a system for controlling fluid simulations by specifying keyframes [Treuille et al. 2003], and Barnat et al. introduced a system for controlling the density of fluids in an example-based manner [Barnat et al. 2011].

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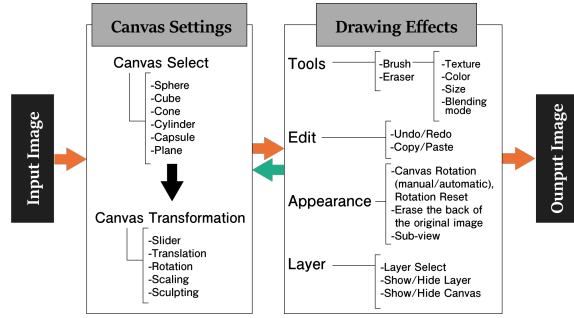


Figure 2: SpiCa framework. Users create flow effects by iterating two processes: setting up a 3D canvas for the input image and drawing effects on the canvas.

In the field of animation, sketch-based methods have been utilized especially for character animation [Thorne et al. 2004]. Kazi et al. proposed Draco as a system that applies such a sketch-based approach to flow effects creation [Kazi et al. 2014]. Instead of utilizing sketches for individual objects, Draco animates a group of similar objects, named kinetic textures, by following their strokes. However, this system does not support a time-varying velocity field, nor are user-sketched shapes deformed.

However, Xing et al. developed a system called Energy-Brushes [Xing et al. 2016], which allows users to design time-varying effect animations from their strokes by incorporating physics-based simulations into the sketch-based interface. User sketches, called energy brushes, act as control gestures for flow particles, and the local velocity field of each flow particle enables simultaneous control of the detailed behavior and coarse structure of flow effects.

To summarize, sketch-based interfaces allow for simpler and more intuitive interaction than physics-based methods, and thus, can be regarded as a powerful means for supporting the design of flow effects. However, most of the common sketch-based interfaces use 2D models because extension to 3D scenes brings ambiguity in interpretation of sketch inputs.

SpiCa, proposed in this paper, provides a novel drawing interface for creating stereoscopic flow effects with 3D pottery wheel-type transparent canvases, on which the sketch inputs can be projected as proper 3D strokes. Limited to effects in character illustrations, SpiCa offers a variety of drawing functions on the canvases.

3 PROPOSED INTERFACE

Fig. 2 shows the SpiCa framework, which takes a transparent image of a character illustration as input and then allows users to iterate two processes: setting up a 3D canvas for the input image and drawing flow effects on the canvas. The actual look and feel of the interface can be seen in the accompanying video.

3.1 Canvas Settings

One approach to designing flow effects in character illustrations is to draw the path of flow effects on a surface with a simple shape, as shown in Fig. 1, to beautify the path of flow effects in an organized composition. General-purpose sketch-based methods can only draw 2D strokes and cannot create complex 3D effects, but SpiCa makes it easy through a 3D transparent canvas.

Fig. 3 is a screenshot of the canvas setting window in SpiCa. The user first selects a canvas shape from the list of sphere, cube, cone,

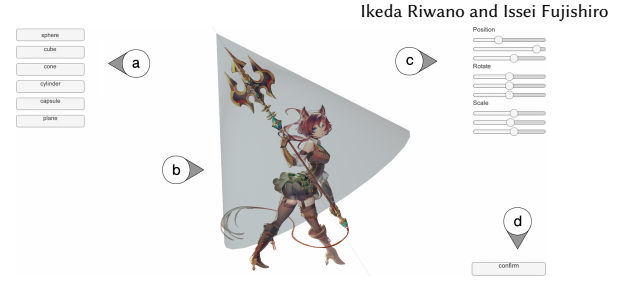


Figure 3: SpiCa's canvas setting window. (a) Canvas shape selection panel, (b) Drag-based transformation area, (c) Slider-based transformation area, (d) Canvas confirmation button.

cylinder, capsule, and plane for the input image. Then, through direct dragging or sliders, translation, rotation and scaling are applied to fit the canvas to the composition of the input image.

The user can also transform the canvas by sketch-based sculpting. When the user confirms shape of their canvas, the system transfers them to the drawing effects window.



Figure 4: SpiCa's drawing effects window. (a) Drawing area, (b) Color picker, (c) Tool select panel, (d) Brush tool panel, (e) Layer panel, (f) Sub-view, (g) Appearance panel, (h) Canvas resetting button.

3.2 Drawing Effects

Fig. 4 is a screenshot of the drawing effects window of SpiCa. In the drawing area, the user can move their viewpoint with a mouse and draw strokes with a pressure-sensitive stylus pen and tablet. The user selects a tool, a brush or an eraser, and paints on the set canvas. Each tool casts a ray from the viewpoint to the clicked position, and drawing is executed when the ray hits the canvas. The brushes are implemented in 3D and are rendered as local solids along the canvas. Various 3D flow effects can be designed by setting colors with the color picker and selecting textures with the brush tool panel.

The scene can be revolved, but the output image is always displayed in the sub-view. The canvas is implemented with the ability to revolve around its axis. This makes it easy to draw 3D strokes such as spirals, as shown in Fig. 5.

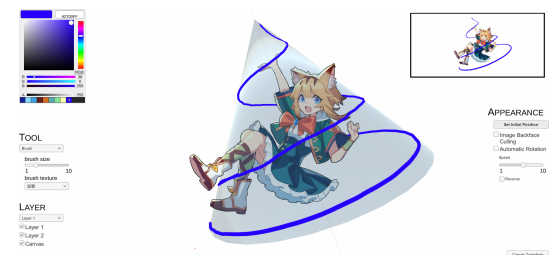


Figure 5: A spiral stroke sketched by automatically revolving the canvas.

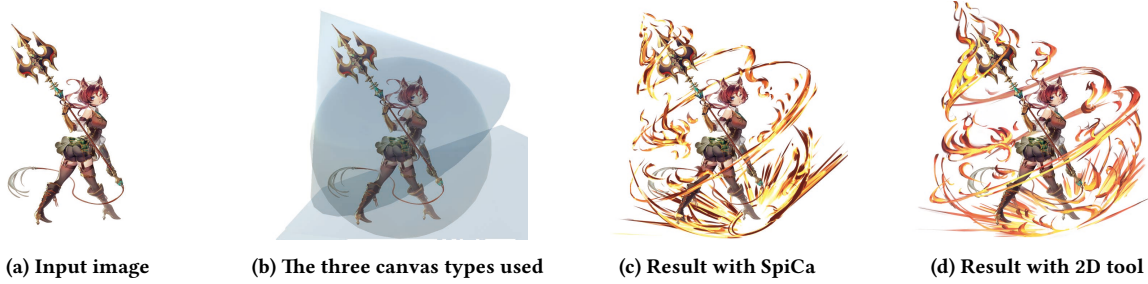


Figure 6: An example executed with SpiCa. An input transparent image is shown in (a). Using three canvas types in (b), the drawing result in (c) was created in 34 min and 50 sec. For comparison, a similar flow effect in (d) created with an existing 2D tool in 50 min and 16 sec.



(a) A canvas created by sculpting (b) Result with sculpting

Figure 7: An execution example using canvas sculpting. The canvas in (a) was used to obtain the drawing result in (b). It took 35 minutes.

With familiarity, it is possible to hide the canvas and draw effects on the canvas even when it is hidden. As the drawing progresses, due to the psychological nature of human vision, the drawn effects allow the user to roughly locate the canvas. In such cases, hiding the canvas enables the user to work with a clearer output image.

3.3 Canvas Resettings

The user can return to the canvas setting window from the drawing effects window. Even after a used canvas is switched, the generated flow effects are retained without any deformation. This makes it possible to readjust the canvas and to work with multiple canvases.

4 RESULTS

We implemented SpiCa using Unity 2019.4.13f1 on a 64-bit Windows 10 (Intel Core i9-10920X 3.50 GHz, RAM 64 GB, GPU NVIDIA GeForce RTX 2070 SUPER).

Fig. 6 shows an example created by the first author. Using the transparent image of the character illustration in Fig. 6(a) as input, the three types of canvases in Fig. 6(b) were used in sequence. The final result with SpiCa is shown in Fig. 6(c). As a control, we created a similar flow effect in Fig. 6(d) using an existing 2D tool under the following conditions:

- Use CLIP STUDIO Paint
- Use a tablet and a stylus pen
- Limit the color mode used to normal and additive.

Compared with the 2D tool, the drawing time with SpiCa was reduced by 31%. Drawing flow effects with the 2D tool tends to cause a 3D collapse in wrap-around areas, but SpiCa enabled the user to create flow effects smoothly. However, SpiCa partially underperformed the existing tool in terms of detail quality.

Fig. 7 shows another example using canvas sculpting. In this example, a sphere-shaped canvas was transformed into a droplet as in Fig. 7(a), to generate the result in Fig. 7(b).

5 USER EVALUATION

To verify the usefulness of SpiCa, we conducted a user evaluation with four participants (P1, P2, P3, and P4). Two women in their 20s and two men in their 20s who all have experience using 2D illustration tools participated in the evaluation to examine the benefits of SpiCa in reducing the drawing time and workload compared to the same existing 2D tool. We also compared the flow effects created with SpiCa and the 2D tool to evaluate the usability and design quality.

First, we provided each participant with a 10-min tutorial to learn how to operate SpiCa. Next, the participants were assigned the task to add to the prepared input images three types of flow effects from (a) to (c) (cone-shaped flame effect, sphere-shaped water effect, cylinder-shaped wind effect) shown in Fig. 1 using SpiCa or the 2D tool. The conditions with the 2D tool were the same as shown in Section 4. For the quantitative evaluation, we measured the drawing time, while for the qualitative evaluation, we used a questionnaire about SpiCa and NASA-TLX [Hart 2006], which was to evaluate the workload.

All the results created with SpiCa and the comparison with those with the 2D tool are shown in Supplement S-1. We confirmed that the participants were able to create wrap-around flow effects in their own expressions.

5.1 Time Measurement Evaluation

Table 1 demonstrates the difference in the drawing time with SpiCa and the 2D tool. The difference is displayed as a percentage of the drawing time using the 2D tool. Negative values indicate that SpiCa decreased the drawing time, while positive values indicate that SpiCa increased it. The absolute values of the drawing time are shown in Supplement S-1.

The table illustrates that there were individual differences in reducing the effect drawing time with SpiCa. Only P1 and P2 took less time with SpiCa for all the effects. The reason might be that P1 and P2 used SpiCa before the 2D tool. In this evaluation, the participants created the same flow effects using SpiCa or the 2D tool; thus, it is likely that once their target image had been finished in either task, the other task became more efficient. Furthermore, the participants were accustomed to painting with the 2D tool but had

Table 1: Difference in drawing time between SpiCa and the existing 2D tool for flow effects from (a) to (c) in Figure 1 (Percentages based on drawing time with the existing tool). Negative values mean that SpiCa decreased the time, while positive values mean that SpiCa increased the time.

Participants	Effects	(a)	(b)	(c)	Mean (SD)
P1		-44%	-40%	-60%	-48% (11%)
P2		-47%	-61%	-62%	-56% (9%)
P3		+111%	+216%	+66%	+131% (77%)
P4		+193%	+185%	+229%	+202% (23%)
Mean (SD)		+53% (119%)	+75% (146%)	+43% (137%)	

never used SpiCa; therefore, so they were not familiar with using it. For these reasons, the results for efficiency in creating flow effects with SpiCa were not argued to be significant.

5.2 NASA-TLX Evaluation

The participants evaluated six workloads for three different flow effects, using SpiCa or the existing tool, following the NASA-TLX [Hart 2006] scale. The evaluation values were obtained using RTLX, which calculates a simple mean value from six different subscales. Table 2 demonstrates the difference in values when each participant created each effect using SpiCa or the existing tool. Negative values indicate that the workload was decreased with SpiCa, while positive values indicate that it was increased with SpiCa. All the RTLX evaluations by each participant are available in Supplement S-2.

The SpiCa workloads of P1, P2, and P3 were smaller in all the cases. In particular, as P1 had the least illustration experience, SpiCa's impact on reducing his workload was evident. In contrast, the workload of the experienced participant P4 was not able to be reduced.

Three out of four participants reported smaller values with SpiCa, suggesting that SpiCa's ability to easily draw stereoscopic flow effects without having to pay attention to 3D consistency reduces artist's mental and physical workloads.

5.3 Questionnaire Evaluation and Comments

After all the flow effects were created, the participants were asked to answer a four-item questionnaire on a five-point scale (1, not at all satisfied, to 5, very satisfied). Table 3 shows the questionnaire items and responses. From the results, it is clear that SpiCa can easily produce high-quality flow effects that satisfy the four participants, compared to the 2D tool. The detailed responses and full text of open-ended comments are available in Supplement S-3.

Table 2: Difference in the total workload estimated by NASA-TLX between SpiCa and the existing 2D tool for flow effects from (a) to (c) in Figure 1. Negative values mean that SpiCa decreased the workload, while positive values mean that SpiCa increased the workload.

Participants	Effects	(a)	(b)	(c)	Mean (SD)
P1		-49.80	-11.23	-26.54	-29.19 (19.42)
P2		-4.44	-17.38	-15.56	-12.46 (7.01)
P3		-5.94	-0.87	-30.17	-12.32 (15.66)
P4		+13.67	0.95	+13.28	+8.67 (8.33)
Mean (SD)		-0.31 (1.24)	-0.41 (2.24)	-0.70 (1.92)	

Table 3: Questionnaire items and responses.

#	Questionnaire items	Mean (SD)
1	Do you think SpiCa is easy to use for creating effects compared to the existing 2D tool?	4.25 (0.96)
2	Do you think it was easy to understand and remember how to use SpiCa?	4.75 (0.50)
3	Do you think your intended design was successfully created with SpiCa compared to the existing tool?	4.25 (0.50)
4	Are you satisfied with the quality of results with SpiCa compared to the existing tool?	4.50 (0.58)

Producing flow effects that surround characters is difficult, especially for beginners unfamiliar with drawing flow effects. Overall, the comments were positive about SpiCa's ability to assist in 3D effect drawing. In addition, some participants reported that drawing flow effects with SpiCa was less stressful and more intuitive.

6 CONCLUDING NOTES

In this paper, we proposed an interface named SpiCa for drawing stereoscopic flow effects on character illustrations using 3D transparent surface of revolution canvases. Through the user evaluation, we confirmed that SpiCa enabled users to create such effects more easily and effectively than an existing 2D illustration tool, and reduced their workload.

However, the conditions and scale of this evaluation did not result in a significant improvement in the efficiency of the design of flow effects. For more efficient creation of such effects, we need to improve the usability of SpiCa as a drawing tool. The comments in Supplement S-3 call for improved customizability and performance for drawing details. Other future work includes extending the effect transformation and enriching composite of the style, perspective, and lighting with input images. At present, we are developing SpiCa as a system for static illustration, but we plan to extend it to a flow effect animation system by employing a method that combines sketch input and physical simulation.

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